

Problem Sheet 4

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1. If DPLL always chooses to assign 0 to a decision variable before assigning 1 (if needed) to a variable, then show that DPLL will never need to backtrack when given a Horn formula encoded in CNF as input.
2. Let's try to solve the HornSAT problem using DPLL. We can convert each Horn clause into a CNF clause by simply rewriting $(a \rightarrow b)$ as $(\neg a \vee b)$, which preserves the semantics of the formula. The resultant formula is in CNF.

Suppose our version of DPLL always assigns 1 to a decision variable before assigning 0 (if needed). How many backtrackings are needed if we run this version of DPLL on the (CNF-ised version of) the following Horn formulae, assuming DPLL always chooses the unassigned variable with the smallest subscript when choosing a decision variable?

(a)

$$\left(\left(\bigwedge_{i=0}^{n-1} x_i \right) \rightarrow x_n \right) \wedge \bigwedge_{i=0}^{n-1} (x_n \rightarrow x_i) \wedge \left(\left(\bigwedge_{i=0}^n x_i \right) \rightarrow \perp \right)$$

(b)

$$\bigwedge_{i=0}^{n-1} ((x_i \rightarrow x_{n+i}) \wedge (x_{n+i} \rightarrow x_i) \wedge (x_i \wedge x_{n+i} \rightarrow \perp))$$

3. Assume that a 2-CNF formula is input to DPLL. Prove that every clause that is learnt is either empty or a singleton.
4. Let us suppose we run DPLL on an unsatisfiable formula. Give a linear time algorithm in terms of the number of steps in the run to generate resolution proof of unsatisfiability from the run of DPLL.
5. Consider the following CNF formula:

$$\begin{aligned} &(\neg p_1 \vee p_2 \vee p_3) \wedge (\neg p_1 \vee p_3 \vee p_9) \\ &\quad \wedge (\neg p_2 \vee \neg p_3 \vee p_4) \\ &\quad \wedge (\neg p_4 \vee p_5) \\ &\quad \wedge (\neg p_4 \vee p_6 \vee \neg p_8) \\ &\quad \wedge (\neg p_5 \vee \neg p_6) \\ &\quad \wedge (p_7 \vee p_1 \vee \neg p_{10}) \\ &\quad \wedge (p_1 \vee p_8) \\ &\quad \wedge (\neg p_7 \vee \neg p_8) \end{aligned}$$

- (a) Check if this CNF formula is satisfiable using the DPLL method.
- (b) Assume $p_9 = \perp$ and $p_{10} = \top$. Now, check if we can find a solution for the above formula using the DPLL method.
6. Consider the formula $\varphi = \forall x \exists y R(x, y) \wedge \exists y \forall x \neg R(x, y)$. Show that φ is satisfiable over a structure whose universe is infinite and countable.
What property should R satisfy so that φ is not satisfiable over any structure with infinite and countable universe?
7. Consider the signature τ that has the binary functions $+, \times$. Let $\mathcal{N}$ be the structure over τ having as universe the set $\mathbb{N}$ of natural numbers and which interprets $+, \times$ in the usual way. Construct FO formulae $\text{Zero}(x)$, $\text{One}(x)$, $\text{Even}(x)$, $\text{Odd}(x)$ and $\text{Prime}(x)$ using τ such that
- For any $a \in \mathbb{N}$, $\mathcal{N} \models \text{Zero}(a)$ iff a is zero.
 - For any $a \in \mathbb{N}$, $\mathcal{N} \models \text{One}(a)$ iff a is one.
 - For any $a \in \mathbb{N}$, $\mathcal{N} \models \text{Even}(a)$ iff a is even.
 - For any $a \in \mathbb{N}$, $\mathcal{N} \models \text{Odd}(a)$ iff a is odd.
 - For any $a \in \mathbb{N}$, $\mathcal{N} \models \text{Prime}(a)$ iff a is prime.

Goldbach's conjecture says that every even integer greater than 2 is the sum of two primes. Whether or not this is true is an open question in number theory. State Goldbach's conjecture as a FO-sentence over τ .

8. Skolem Arithmetic is the first-order theory of the natural numbers with multiplication. Formulae in Skolem Arithmetic (SA) use first-order variables $x, y, \dots$ that range over natural numbers, $\mathbb{N} = \{0, 1, 2, \dots\}$ and the ternary multiplication predicate $x * y = z$. This is the only predicate available to you, other than the usual equality predicate ($=$). Moreover, you are free to use constants from $\mathbb{N}$. Here is a well formed sentence in this logic:

$$\forall x \forall y \exists z (x * z = y)$$

Construct formulae in SA which capture the following:

- (a) $\text{Divide}(a, b)$: a divides b .
- (b) $\text{GCD}(a, b, d)$: d is the GCD (greatest common divisor) of a and b .
- (c) $\text{LCM}(a, b, e)$: e is the LCM (least common multiple) of a and b .
- (d) $\text{Prime}(p)$: p is a prime number.
- (e) $\text{k-Product}(b)$: b is the product of k primes, where k is a fixed number.
- (f) $\text{Prime-Power}(b)$: b is a power of some prime.